

Wearable Textile Triboelectric Glove for Grasp Force and Pressure Sensing

Don Perera^{1,2}, Pranavi Vedula², Wenzhuo Wu^{1,2,3,4,5,6}



Edwardson School of
Industrial Engineering

¹Edwardson School of Industrial Engineering

²Flex Laboratory

³Purdue Institute for Integrative Neuroscience

⁴Women's Global Health Institute

⁵Purdue's Institute of Inflammation, Immunology and Infectious Disease

⁶Purdue's Institute for Drug Discovery

INTRODUCTION

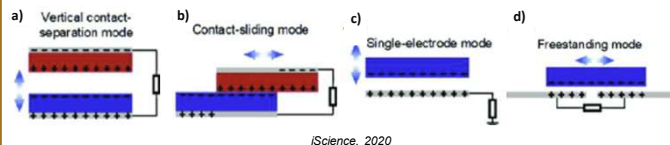
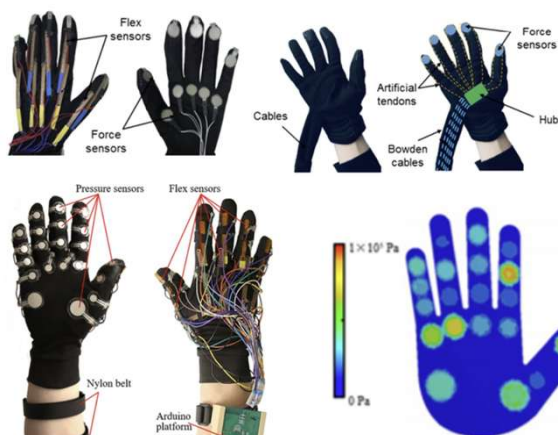


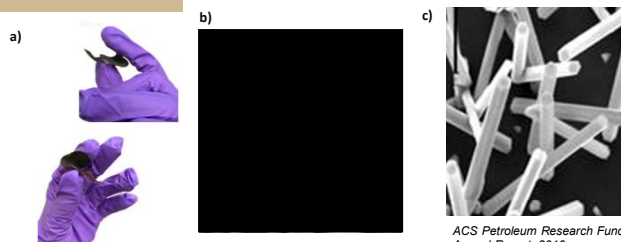
Figure 1. a) Vertical contact separation: two surfaces repeatedly come into contact and separate, generating alternating current due to periodic charge transfer. **b)** Contact-sliding mode: produces signals through relative in-plane motion between two materials, where sliding changes the distribution of surface charges. **c)** Single-electrode mode: operates with only one electrode connected, using the environment or human body as a reference, making it especially suitable for wearable systems. **d)** Freestanding mode: involves a moving charged layer that induces current between electrodes without direct electrical connection.



Advanced Intelligence Systems, 2022

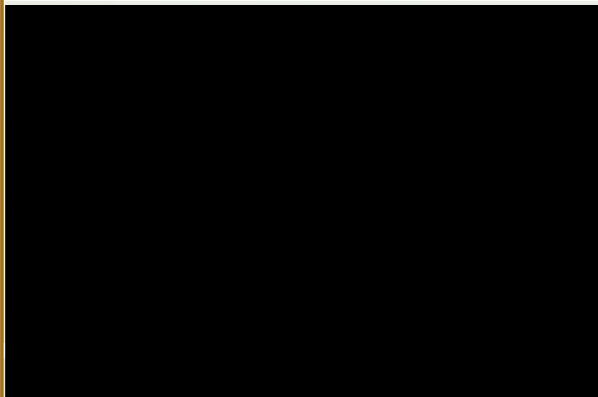
Figure 2. A wearable triboelectric glove is developed to capture grasp force and pressure during object interaction. The generated signals can be used for digital twin applications, where real-world data is collected to train virtual models for robotics and human-machine systems. The flexible, textile-based design also supports ergonomic use, allowing natural hand movement while maintaining high sensing performance.

MATERIAL SELECTION



ACS Petroleum Research Fund
Annual Report, 2016

DEVICE ARCHITECTURE



DEVICE FABRICATION

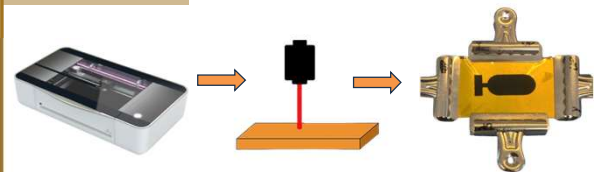
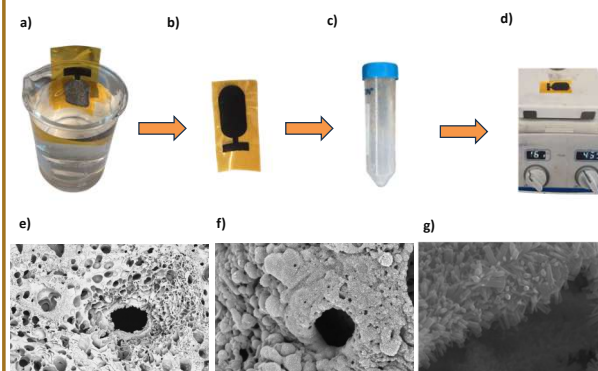


Figure 5. PI Film Fabrication (Fingertip Patterning) : Polyimide (PI) film which serves as the base material for the sensing regions is cut into the desired fingertip geometries to match the contact areas of the glove



DEVICE PERFORMANCE

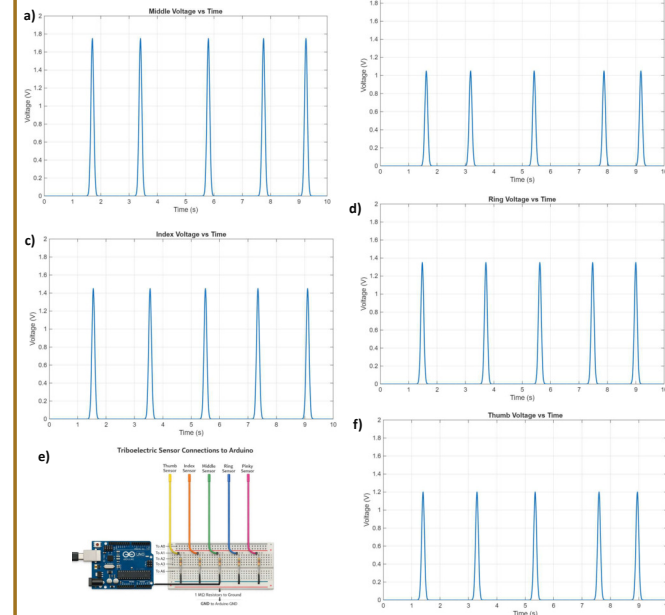


Figure 7. a) Middle finger voltage spikes over time. **b)** Pinky finger voltage spikes over time. **c)** Index finger voltage spikes over time. **d)** Ring finger voltage spikes over time. **e)** Arduino and triboelectric sensor wiring setup. **f)** Thumb finger voltage spikes over time.

APPLICATIONS

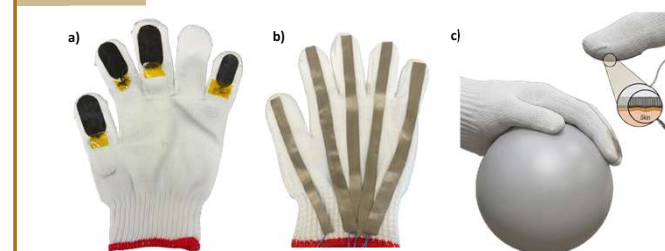


Figure 8. a) LIG-based triboelectric sensors integrated into working glove. **b)** conductive fabric covered wiring inside the glove to reduce noise effects. **c)** Schematic of the glove in potential pressure sensor applications.

REFERENCES

1. Zhang, Tongtong, et al. "Recent Progress in Hybridized Nanogenerators for Energy Scavenging." *iScience*, vol. 23, no. 11, 2020, article 101689.
2. Ozioko, Oliver, and Ravinder S. Dahiya. "Smart Tactile Gloves for Haptic Interaction, Communication, and Rehabilitation." *Advanced Intelligent Systems*, vol. 4, no. 2, 2022, article 2100091.
3. Hahn, Jong-in. "Micro/Nanosopic Investigation of ZnO towards the Development of Next-Generation Separation Platforms." *61st Annual Report on Research*, American Chemical Society Petroleum Research Fund, 2016.